

UNIVERSITI KEBANGSAAN MALAYSIA
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Tunnel Diodes, Resonant-Tunnelling Diodes and Tunnel Transistors

KKKC6353 Nanoelectronic Devices — Lecture 3, Document 2 of 3

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How to use this document

Document 1 showed that electrons pass through thin walls, and that the chance of getting through depends enormously on the wall. This document puts that effect to work in three electronic devices:

- the **tunnel diode**, a p–n junction doped so heavily that its wall is thin enough to tunnel through, which gives a current that *falls* as the voltage rises;
- the **resonant-tunnelling diode**, two walls with a quantum well between them, which lets electrons through only at one energy and makes the fastest electronic oscillators ever built;
- the **tunnel transistor**, or tunnel FET, an attempt to make transistors that switch with less voltage, and so use less power, than any ordinary transistor can.

You will need the idea of energy bands from Lecture 1: electrons in a semiconductor sit in a lower, nearly full **valence band** and can move freely in an upper **conduction band**, with a gap between them where they cannot be. The **Fermi level** marks how high the electrons are filled. Doping with donors pushes it up towards the conduction band (n-type); doping with acceptors pulls it down towards the valence band (p-type).

As before, each section opens with a blue **Key ideas** box and closes with **check questions**, and the green **Going further** boxes hold the optional mathematics.

1 The tunnel diode

> **Key ideas**
>
> An ordinary p–n junction has a wide region in the middle, far too thick to tunnel through.
>
> Dope both sides very heavily and that region shrinks to about 10 nm. Electrons can then tunnel straight across it.
>
> Leo Esaki discovered this in 1957, and shared the 1973 Nobel Prize in Physics for it.

1.1 A thin wall in a p–n junction

Where n-type and p-type semiconductor meet, electrons and holes near the junction cancel each other out, leaving a **depletion region** with no free carriers. In an ordinary diode this region is hundreds of nanometres to a few micrometres wide. For an electron in the n side's conduction band to reach the p side, it must either climb over the energy step or cross the whole depletion region, a wall far too thick to tunnel through.

The more heavily a junction is doped, the narrower its depletion region. In a **tunnel diode** both sides are doped about a thousand times more heavily than in an ordinary diode, around 10^{19} to 10^{20} dopant atoms per cubic centimetre. The depletion region shrinks to about 10 nm. Now the wall is thin enough that electrons can tunnel straight through it.

Heavy doping does something else too. It pushes the Fermi level right *into* the bands: into the conduction band on the n side and into the valence band on the p side. So the n side has a strip of filled states in its conduction band, and the p side has a strip of *empty* states (holes) at the top of its valence band. These two strips, facing each other across the thin wall, are what make the tunnel diode work.

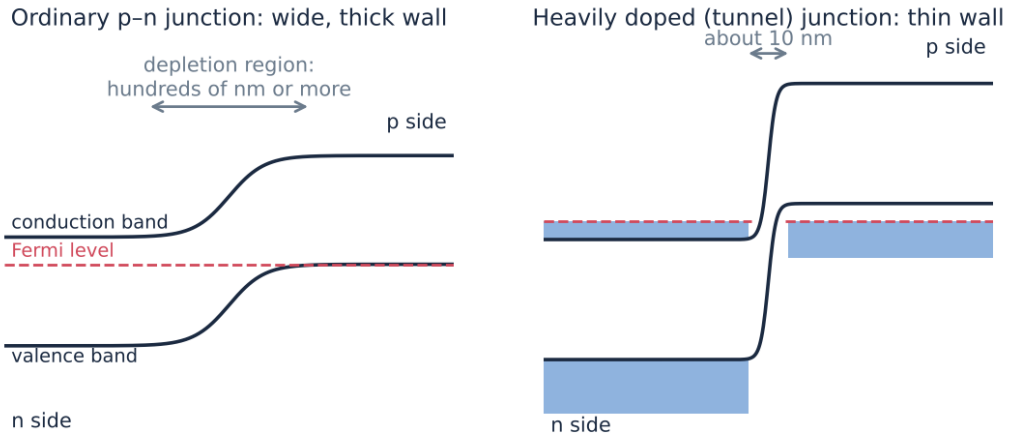


Figure 2.1. *Left: an ordinary p–n junction. The energy step is spread over a wide depletion region, a wall far too thick to tunnel through. Right: a very heavily doped junction. The step is only about 10 nm wide, and the Fermi level (dashed) lies inside the bands; shading shows the filled states. Figure made for this course; schematic.*

1.2 Esaki's discovery

In 1957 Leo Esaki, a young researcher at Tokyo Tsushin Kogyo, the company that soon renamed itself Sony, was studying very heavily doped germanium junctions. He found something no diode had shown before: as he raised the forward voltage, the current first rose, then *fell*, then rose again. He explained it as tunnelling and published it in 1958. In 1973 he shared the Nobel Prize in Physics with Ivar Giaever and Brian Josephson, both of whom studied tunnelling in superconductors. We will meet Josephson's work in Document 3.

Check your understanding

1. Why can electrons not tunnel across an ordinary p–n junction?
2. What two things does very heavy doping do to the junction?

2 Walking along the tunnel diode's curve

> **Key ideas**
>
> Electrons can only tunnel into an empty state at the same energy.
>
> As the forward voltage rises, the filled states on the n side first line up with the empty states on the p side, then slide past them.
>
> So the tunnelling current rises to a peak and then falls. After that the ordinary diode current takes over.

2.1 The rule: filled facing empty

An electron tunnelling across the junction keeps its energy (Document 1). So it can only tunnel into a state *at the same energy* on the other side, and that state must be empty. Tunnelling current flows when filled states on one side face empty states on the other.

The voltage applied to the diode slides the two sides up and down relative to each other. Follow the figure below from (a) to (e).

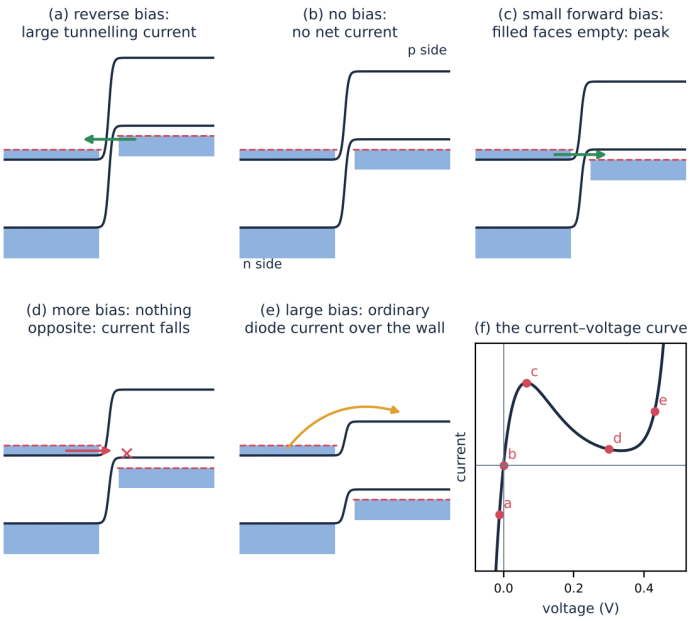


Figure 2.2. *The tunnel diode, step by step. Shading shows filled states; the dashed lines are the Fermi levels on each side. (a) Reverse bias: filled valence states on the p side face empty conduction states on the n side, and electrons tunnel from right to left. (b) No bias: nothing faces anything empty; no net current. (c) Small forward bias: the n side's filled strip faces the p side's empty strip; tunnelling current peaks. (d) More bias: the strips have slid past each other; there is nothing empty opposite the filled states, and the current falls. (e) Large bias: the step is now low enough for electrons to go over it, as in an ordinary diode. (f) The resulting current–voltage curve. Figure made for this course; schematic.*

- **(b) No voltage.** The Fermi levels on the two sides line up. Tunnelling can happen both ways, equally, so there is no net current.
- **(c) Small forward voltage.** The n side's filled conduction-band strip rises to face the p side's empty valence-band strip. Electrons tunnel from left to right, and the current grows quickly. It reaches its **peak** when the overlap is largest, typically at a few tens of millivolts.
- **(d) More voltage.** The n side's filled strip has risen *past* the top of the p side's valence band. Opposite the filled states there is now only the band gap, where there are no states at all. Tunnelling stops, and the current **falls** even though the voltage is rising. It reaches a low point, the **valley**, typically at a few hundred millivolts.
- **(e) Large forward voltage.** The energy step across the junction is now low enough for electrons to go *over* it, as in any ordinary diode. The current rises again, steeply.
- **(a) Reverse voltage.** The p side's filled valence band now faces the n side's empty conduction band. Electrons tunnel from p to n, and a large current flows for even a small reverse voltage. A tunnel diode conducts readily in reverse.

Watch. NJPhysics, *Tunnel diode*, a clear walk through the band diagrams, youtube.com/watch?v=iV0P5siu4cs.

2.2 Three currents added together

The measured current is the sum of three parts. The tunnelling current makes the first hump. The ordinary diode current makes the final rise. In between, a small **excess current** fills in the valley: electrons tunnel through stray states in the band gap created by defects

and the heavy doping. The excess current is why the valley never reaches zero, and keeping it small is a large part of making a good tunnel diode.

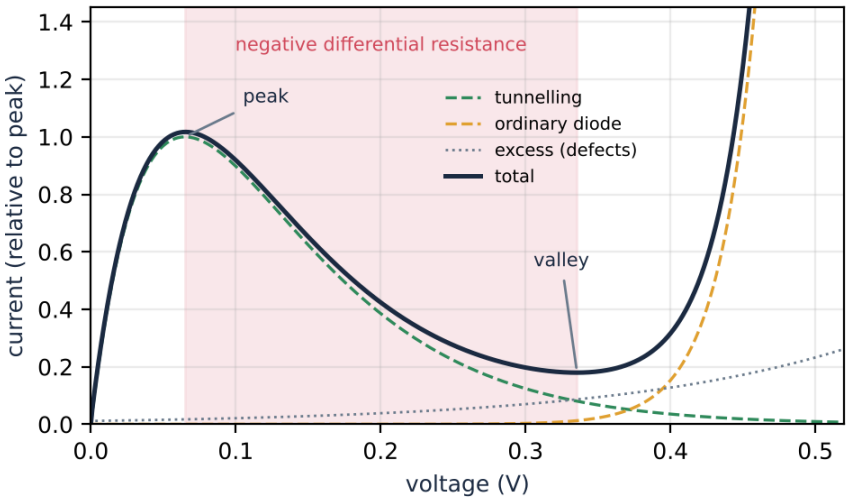


Figure 2.3. *The tunnel diode current (solid) as the sum of a tunnelling current, the ordinary diode current and a small excess current through defects. Between the peak and the valley (shaded) the current falls as the voltage rises: negative differential resistance. Figure made for this course from the model in the Going further box.*

The ratio of the peak current to the valley current, the **peak-to-valley current ratio**, is the usual figure of merit. The larger it is, the more useful the falling part of the curve.

> **Going further: a model of the curve (optional)**
>
> A widely used empirical model writes the current as $I = I_{\text{tun}} + I_{\text{diode}} + I_{\text{excess}}$, with
>
> $$I_{\text{tun}} = \frac{V}{R_0}\exp\left[-\left(\frac{V}{V_0}\right)^{m}\right], \qquad I_{\text{diode}} = I_s\left(e^{qV/k_BT} - 1\right).$$
>
> R_0, V_0 and m (between 1 and 3) are fitted to the device; V_0 is typically 0.1 to 0.5 V. The tunnelling current peaks where $dI_{\text{tun}}/dV = 0$, at
>
> $$V_p = \left(\frac{1}{m}\right)^{1/m} V_0.$$
>
> The figure above uses $m = 1$, so $V_p = V_0 = 65$ mV, which is typical of germanium; it gives a peak-to-valley ratio of about 5.6. The excess current is modelled as $I_v \exp[C(V - V_v)]$ near the valley.

Check your understanding

1. State the rule that decides whether electrons can tunnel across the junction.
2. Explain, using band pictures, why the current *falls* between the peak and the valley.
3. Why does a tunnel diode conduct strongly in reverse, unlike an ordinary diode?

Explore. The animation *Tunnel diode* (faizverse.github.io/nanoelectronic-devices/L3) lets you slide the voltage and watch the bands and the curve move together.

3 What a falling current is good for

> **Key ideas**
>
> A region where current falls as voltage rises is called negative differential resistance.
>
> It can cancel the losses in a resonant circuit, so the circuit oscillates on its own: an oscillator.
>
> Tunnelling is extremely fast, so tunnel diodes oscillate and switch at very high frequencies.

3.1 Negative differential resistance

An ordinary resistor obeys Ohm's law: more voltage, more current. Its *resistance*, the change in voltage divided by the change in current, is positive. Between its peak and its valley a tunnel diode does the opposite: more voltage, *less* current. The change in voltage divided by the change in current is negative. This is called **negative differential resistance**, or NDR.

A common misconception

"Negative resistance means the diode produces power." It does not. The diode still takes power from its battery: the current and voltage are both positive, so their product is positive. "Differential" is the key word. It is only the *changes* that have opposite signs. A small wiggle in voltage produces a wiggle in current the opposite way, and that is what lets the diode feed energy into an oscillation, drawing it from the battery.

3.2 Oscillators

Any resonant circuit, such as a coil and a capacitor, rings when struck, like a bell. Its resistance takes energy out at every cycle, and the ringing dies away. Add a tunnel diode set at a voltage in its falling region, and its negative differential resistance puts energy back in. If it puts in more than the circuit loses, a tiny disturbance grows into a steady oscillation, powered by the battery. That is an **oscillator**, the heart of every radio transmitter and radar.

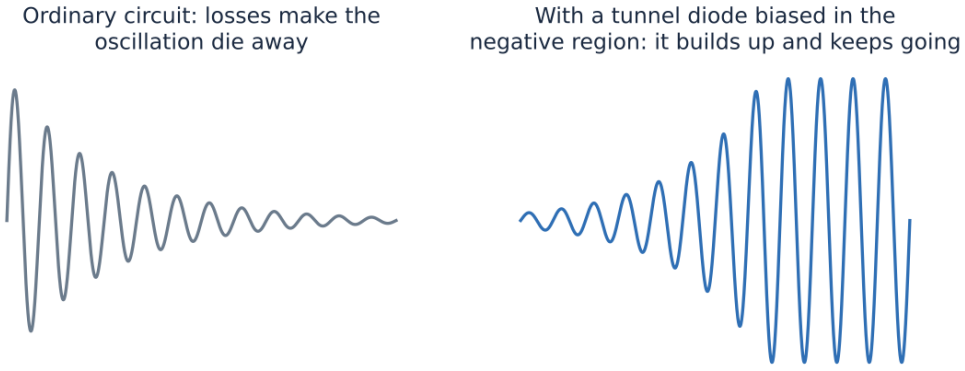


Figure 2.4. *Left: an ordinary resonant circuit rings and dies away because of its losses. Right: with a tunnel diode biased in its negative-resistance region, the losses are cancelled and the oscillation grows to a steady size. Figure made for this course; schematic.*

Tunnelling itself takes almost no time, much less than a picosecond. What limits how fast a tunnel diode can go is the time to charge its own small capacitance through its resistance.

Tunnel diodes were the fastest devices of the early 1960s, used in microwave oscillators, amplifiers and very fast switching circuits.

3.3 Where tunnel diodes went, and where they are now

Transistors soon overtook them. A tunnel diode has only two terminals, so the input and the output share the same pair of wires, which makes circuits awkward. Its current is small, and the falling region spans only a few hundred millivolts. Transistors can be packed by the billion on one chip; tunnel diodes cannot.

But heavily doped tunnel junctions are still at work in an important place: the most efficient **solar cells**, used on almost every satellite, are stacks of two to six cells made of different semiconductors, each absorbing a different part of sunlight. The cells must be connected in series, and a normal p–n junction between them would block the current. Instead they are joined by thin **tunnel junctions**, which let the current pass from one cell to the next with almost no voltage lost. Close relatives, called *backward diodes*, are also used as sensitive microwave detectors.

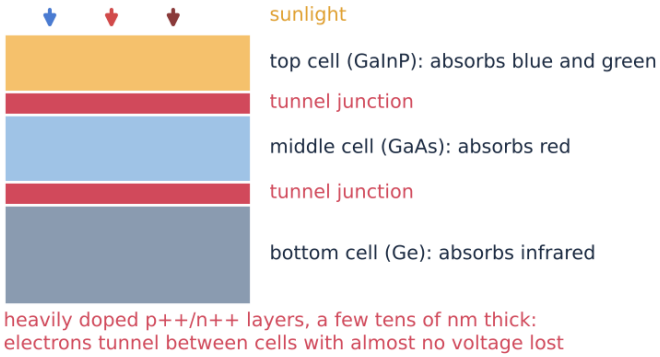


Figure 2.5. *A multi-junction solar cell. Each cell absorbs a different part of sunlight. They are joined by very heavily doped tunnel junctions, through which electrons tunnel from one cell to the next. Figure made for this course; layer thicknesses not to scale.*

Check your understanding

1. What does “negative differential resistance” mean? Does the diode generate power?
2. How can a tunnel diode keep a resonant circuit oscillating?
3. Give two reasons why transistors replaced tunnel diodes in most uses.

4 The resonant-tunnelling diode

> **Key ideas**
>
> Put two very thin walls a few nanometres apart and you have a quantum well between them, with its own allowed energy level.
>
> Electrons with exactly that energy pass through both walls almost perfectly. Others are almost all reflected. This is resonant tunnelling.
>
> A voltage moves the level. When it slides below the incoming electrons, the current drops: negative differential resistance again, but now very fast.

4.1 Two walls make a filter

Lecture 2 showed that a thin layer of GaAs between layers of AlGaAs is a **quantum well**, with only certain allowed energy levels. Make the AlGaAs walls on each side very thin, about 2 nm, and electrons can tunnel in and out of the well. This structure is a **resonant-tunnelling diode**, or RTD.

Something remarkable happens. By Document 1, an electron meeting one 2 nm wall gets through only rarely; meeting two in a row should be rarer still. But if the electron's energy matches the level in the well, the waves reflected back and forth between the two walls add up in step, like a guitar string driven at its own note, and the electron passes through *both* walls almost every time. At every other energy it is almost always reflected. The double wall acts as an extremely sharp **energy filter**.

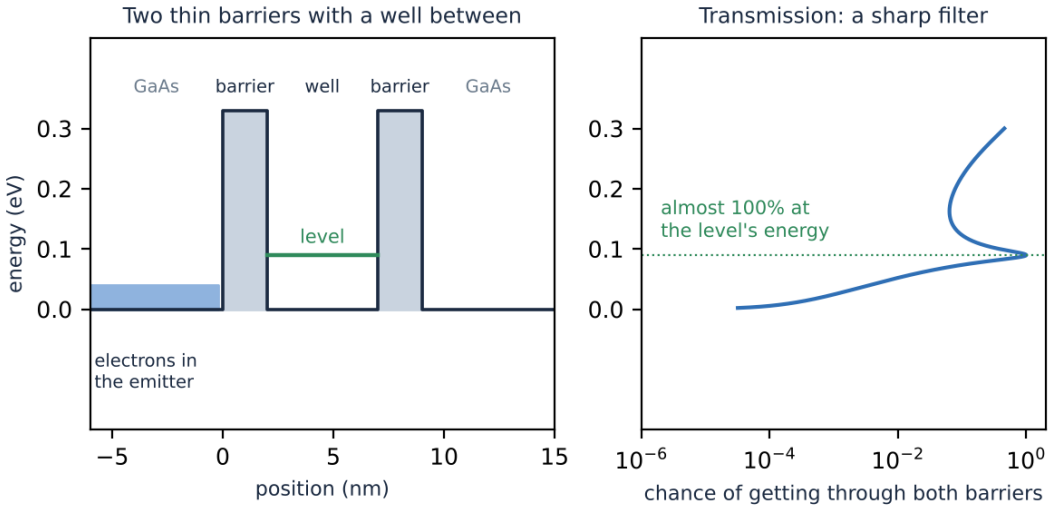


Figure 2.6. *Left: a resonant-tunnelling diode. Two AlGaAs barriers, 2 nm thick and 0.33 eV high, enclose a 5 nm GaAs well with one allowed level (green). Right: the chance of getting through both barriers, against energy. It is close to 100% at the level's energy and tiny elsewhere. Figure made for this course, from an exact calculation with an effective mass of 0.067 m_0.*

Optics has the same effect. Two partly silvered mirrors facing each other, the **Fabry–Pérot** interferometer, transmit light almost perfectly at certain wavelengths and reflect it at all others.

4.2 Moving the level

The electrons arrive from the left-hand GaAs layer, the **emitter**, which is doped so that its conduction band is filled up to a little above its bottom. A voltage across the device tilts the energy picture, pulling the well's level down relative to the emitter.

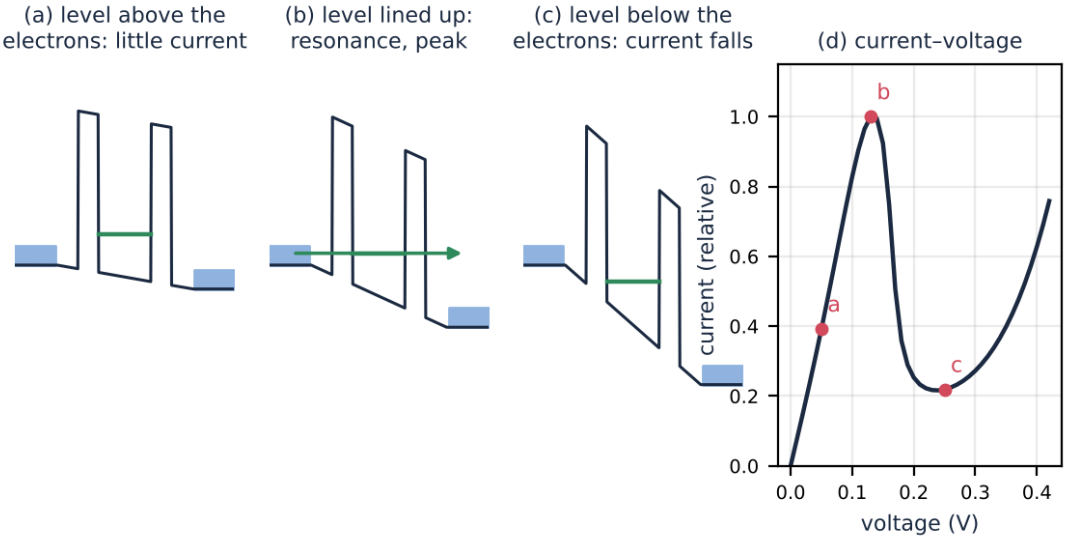


Figure 2.7. *The resonant-tunnelling diode under increasing voltage. (a) The level is above the emitter's electrons: little current. (b) The level lines up with the electrons: they pass through by resonance, and the current peaks. (c) The level has dropped below the bottom of the emitter's band: no electron has the right energy, and the current falls. (d) The calculated current–voltage curve. Figure made for this course, calculated for the structure in the previous figure at room temperature.*

- **(a)** At small voltage the level sits above the emitter's electrons. Few electrons have the right energy, and the current is small.
- **(b)** As the voltage rises, the level comes down into line with the emitter's electrons. Many pass through, and the current reaches a **peak**.
- **(c)** More voltage pulls the level *below* the bottom of the emitter's conduction band. No electron in the emitter has an energy that low, so resonance stops and the current falls to a **valley**.

At still higher voltages the current rises again as electrons find ways over the barriers or through higher levels. The result looks much like the tunnel diode's curve, but for a different reason: here the filter slides out of line with the electrons, rather than filled states sliding past empty ones. For the structure in the figure, our calculation gives a peak at about 0.13 V and a peak-to-valley ratio of about 4.6 at room temperature.

The idea was proposed by Raphael Tsu and Leo Esaki in 1973, and the first device was demonstrated by Chang, Esaki and Tsu in 1974. It was one of the first devices made possible by **molecular beam epitaxy** (Lecture 2, Document 3), which can grow layers a few atoms thick.

> **Going further: where the peak comes from (optional)**
> For a well of width $L = 5$ nm and walls 0.33 eV high in GaAs, the allowed level sits at $E_1 \approx 90$ meV above the bottom of the well (lower than the perfect-walls value of 225 meV because the wave leaks into the walls; see Lecture 2, Document 1).

> The applied voltage V drops across the whole active region, so the level at the middle of the well moves down by roughly $qV/2$. Resonance is lost when the level falls below the bottom of the emitter's band, $E_1 - qV/2 \approx 0$, so the peak lies a little below $V \approx 2E_1/q \approx 0.18$ V. The exact calculation, which follows the level through the emitter's spread of electron energies, puts it at 0.13 V.
>
> The current is $J \propto \int T(E,V)\, N(E,V)\, dE$, where T is the transmission through the structure and N counts the electrons available at each energy (the Tsu–Esaki formula). The figures in this section were calculated this way.

4.3 The fastest electronic oscillators

Electrons pass through an RTD in a fraction of a picosecond, and the device can be made very small, so its capacitance is tiny. RTDs therefore oscillate at frequencies no other electronic device can reach at room temperature. In 1997, Reddy and co-workers at the University of California, Santa Barbara, made RTD oscillators running at 650 GHz. In 2017, Asada's group at the Tokyo Institute of Technology reached **1.98 THz**, nearly two million million cycles per second.

This is the **terahertz** range, between microwaves and infrared light, long called the "terahertz gap" because it is so hard to generate. Terahertz waves pass through clothing, paper and plastics but not metal or water, and many molecules absorb them at characteristic frequencies. RTD sources are being developed for security imaging, gas sensing, inspection of packaged goods and very-high-speed short-range wireless links of the kind proposed for future "6G" networks. Their weakness is power: a single RTD gives only microwatts at the highest frequencies, so arrays and better antennas are an active research area.

A common misconception

"The 600 GHz tunnel diode." You may meet this in older notes. The record oscillators of the late 1990s and since are resonant-tunnelling diodes, not Esaki tunnel diodes. Both show negative differential resistance through tunnelling, but by different mechanisms, and RTDs are much faster.

Check your understanding

1. Why does an electron pass through two thin barriers almost perfectly at one energy, but hardly at all at others?
2. Using the energy level in the well, explain why an RTD's current falls after its peak.
3. Why can RTDs oscillate at terahertz frequencies, and what limits their use?

Explore. The animation *Resonant-tunnelling diode* (faizverse.github.io/nanoelectronic-devices/L3) lets you change the voltage and watch the level, the transmission peak and the current.

5 The tunnel transistor

> **Key ideas**
>
> An ordinary transistor switches by raising and lowering an energy barrier. At room temperature it needs at least 60 mV of gate voltage for every tenfold change in current.
>
> This limit stops us lowering the supply voltage, and so the power, of chips.
>
> A tunnel FET switches by turning tunnelling on and off instead, and in principle can beat the 60 mV limit. It is still a research device.

5.1 The 60 millivolt problem

Lecture 1, Document 2 explained why an ordinary transistor, a MOSFET, cannot switch off sharply. Electrons pass from source to drain by going *over* an energy barrier that the gate raises and lowers. The electrons in the source do not all have the same energy: like the molecules of a gas, most have little and a few have a lot. These few "hot" electrons get over the barrier even when the transistor is meant to be off.

Lowering the barrier lets more electrons over, but only gradually. At room temperature, the gate voltage must change by at least **60 millivolts** for each tenfold change in current. To turn a transistor from fully on to properly off, say a factor of a million in current, needs at least six times 60 mV, plus a margin. This is why the supply voltage of chips has hardly fallen for about twenty years, staying around 0.7 to 1 V, and since power grows with the square of voltage, it is a major reason why chips run hot.

5.2 Switching by tunnelling

A **tunnel FET** (TFET) uses the same gate, but a different source. The source is doped p-type instead of n-type, so the electrons it supplies come from its *valence band*. With the gate off, the channel's conduction band sits above the source's filled valence states, and nothing can tunnel. Turn the gate on, and it pulls the channel's bands down until its conduction band drops below the top of the source's valence band. A narrow window opens, and electrons tunnel through the thin wall between source and channel.

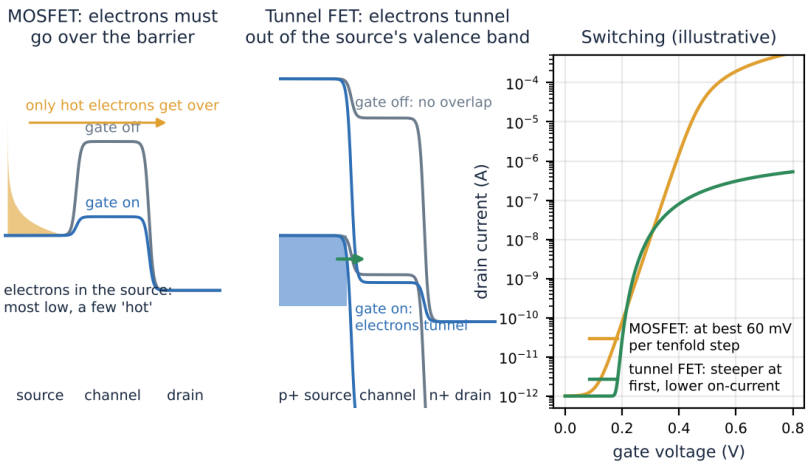


Figure 2.8. *Left: a MOSFET switches by letting hot electrons over a barrier, so at best its current changes tenfold per 60 mV at room temperature. Middle: a tunnel FET's gate pulls the channel's conduction band below the source's filled valence band, and electrons tunnel across. Right: illustrative switching curves. Figure made for this course; schematic.*

The key is that the window cuts off the hot electrons. Electrons above the top of the source's valence band have nothing to tunnel into, and electrons below the bottom of the channel's conduction band have nowhere to go. Only those in the window flow, and the window opens sharply. So a TFET is not bound by the 60 mV limit, and in principle could switch fully with a much smaller voltage.

5.3 Where tunnel FETs stand

Laboratory TFETs have switched more steeply than 60 mV per tenfold step, but usually only over a small range of very low currents. Their main problem is **on-current**: tunnelling through even a thin wall lets fewer electrons through than going over a lowered barrier, so a TFET that is fully on carries far less current than a MOSFET of the same size, and circuits made from it are slow. Defects in the junction also add unwanted current that spoils the steep switching.

Researchers attack this by using materials with small band gaps and light electrons at the junction (germanium, III–V compounds such as InAs and GaSb, and two-dimensional materials), and by surrounding the channel with the gate as in the nanosheet transistors of Lecture 1. As of 2026, no tunnel FET is used in mainstream chips. It remains one of several "steep-slope" ideas, alongside negative-capacitance FETs, that the industry is watching.

> **Going further: where 60 mV comes from (optional)**
> In a MOSFET the number of electrons with enough energy to cross a barrier of height ϕ follows the Boltzmann tail, $n \propto e^{-q\phi/k_BT}$. If the gate lowers the barrier one-for-one with its voltage, the current rises by a factor of 10 when $q\,\Delta V = k_BT\ln 10$:
>
> $$SS = \ln(10)\,\frac{k_BT}{q} \approx 2.30 \times 25.9\ \text{mV} \approx 60\ \text{mV per decade at 300 K.}$$
>
> In a real transistor the gate does not control the barrier perfectly, so SS is larger (Lecture 1, Document 2). In a tunnel FET, the current is set by the tunnelling probability and the width of the energy window, neither of which follows the Boltzmann tail, so SS can in principle be smaller than 60 mV per decade.

Check your understanding

1. Why can an ordinary transistor not change its current by more than ten times for every 60 mV on its gate at room temperature?
2. How does a tunnel FET avoid this limit?
3. What is the main problem holding tunnel FETs back?

Explore. The animation *Tunnel FET versus MOSFET* (faizverse.github.io/nanoelectronic-devices/L3) lets you move the gate voltage and compare the two switches.

6 How these devices are made

> **Key ideas**
>
> All three devices depend on layers or junctions only a few nanometres thick.
>
> Early tunnel diodes were made by melting a dot of metal into a heavily doped crystal; RTDs need atomically precise layer-by-layer growth.
>
> Tunnel FETs are built on the same production lines as ordinary transistors.

Tunnel diodes. Esaki's germanium diodes, and most commercial tunnel diodes of the 1960s, were *alloy junctions*. A small pellet of metal containing the dopant was placed on a heavily doped wafer and heated until it melted into the surface. On cooling, a thin, very heavily doped regrown layer formed an abrupt junction. Modern tunnel junctions, such as those inside multi-junction solar cells, are grown by epitaxy, with very heavily doped layers each a few tens of nanometres thick.

Resonant-tunnelling diodes. The barriers and the well must each be a few nanometres thick, and their thickness controls the current exponentially, so they are grown by **molecular beam epitaxy** or **metal-organic vapour-phase epitaxy** (Lecture 2, Document 3), which deposit one atomic layer at a time. The GaAs/AlGaAs system was the first; the fastest modern RTDs use InGaAs wells with AlAs barriers on indium phosphide (InP) substrates. Electron-beam lithography then defines contacts and devices less than a micrometre across, and an antenna is formed on the chip to send the terahertz waves out.

Tunnel FETs. These use the ordinary transistor process of Lecture 1, with one important change: the source and drain are doped with opposite types. The hard part is making the source–channel junction both very abrupt and free of defects, especially when a different material such as germanium is used for the source.

Check your understanding

1. Why must RTD layers be grown one atomic layer at a time?
2. What is the one essential difference between how a tunnel FET and an ordinary MOSFET are made?

7 Summary, and what comes next

1. **A tunnel diode** is a p–n junction doped so heavily that its wall is about 10 nm thick. Electrons tunnel from filled states into empty states facing them.
2. **Its current rises, falls and rises again.** The fall, negative differential resistance, happens because the filled states slide past the empty ones.
3. **Negative differential resistance makes oscillators.** It cancels the losses in a resonant circuit. Tunnel junctions live on today inside the best solar cells.
4. **A resonant-tunnelling diode** is two thin walls around a quantum well. It lets electrons through only when their energy matches the well's level, and oscillates at up to about 2 THz.
5. **Ordinary transistors need at least 60 mV per tenfold change in current**, because they switch by letting hot electrons over a barrier.

6. **A tunnel FET** switches by opening a tunnelling window instead and could beat that limit, but its on-current is still too low for mainstream use.

Where Document 3 begins. All the devices here pass huge numbers of electrons at once. Document 3 asks what happens when electrons tunnel onto an island so small that adding a *single* electron changes its energy noticeably. The answer is a transistor that controls current one electron at a time, and, with superconductors, the circuits behind today's quantum computers.

Words to know

Word	Plain meaning
Degenerate doping	Doping so heavy that the Fermi level lies inside a band
Tunnel diode (Esaki diode)	A very heavily doped p–n junction whose current is carried by tunnelling at low voltage
Negative differential resistance	A region where current falls as voltage rises
Peak-to-valley current ratio	Peak current divided by valley current; the larger the better
Excess current	Unwanted current through defect states in the band gap
Resonant tunnelling	Near-perfect tunnelling through two barriers at the energy of the level between them
Resonant-tunnelling diode (RTD)	A double-barrier quantum well device with negative differential resistance
Terahertz	Frequencies of about 0.1 to 10 THz, between microwaves and infrared
Subthreshold swing	Gate voltage needed for a tenfold change in current; at least 60 mV in a MOSFET at room temperature
Tunnel FET (TFET)	A transistor that switches by turning band-to-band tunnelling on and off

Sources and further reading

Gentle introductions

- NJPhysics, *Tunnel diode*, youtube.com/watch?v=iV0P5siu4cs.
- Physics and Radio-Electronics, *How a tunnel diode works*, physics-and-radio-electronics.com.

Textbooks

- S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*, 3rd ed., Wiley, 2007. Chapter 8 (tunnel diode and resonant-tunnelling diode).
- J. H. Davies, *The Physics of Low-Dimensional Semiconductors*, Cambridge University Press, 1998. Chapter 5 (tunnelling and resonant tunnelling).

Original work and reviews

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- L. L. Chang, L. Esaki and R. Tsu, “Resonant tunneling in semiconductor double barriers”, *Applied Physics Letters* 24, 593 (1974).
- M. Reddy et al., “Monolithic Schottky-collector resonant tunnel diode oscillator arrays to 650 GHz”, *IEEE Electron Device Letters* 18, 218 (1997).
- R. Izumi, S. Suzuki and M. Asada, “1.98 THz resonant-tunneling-diode oscillator with reduced conduction loss by thick antenna electrode”, 42nd International Conference on Infrared, Millimeter and Terahertz Waves (IRMMW-THz), 2017.
- M. Asada and S. Suzuki, “Terahertz emitter using resonant-tunneling diode and applications”, *Sensors* 21, 1384 (2021). An open-access review.
- A. M. Ionescu and H. Riel, “Tunnel field-effect transistors as energy-efficient electronic switches”, *Nature* 479, 329 (2011).

Credit. Sections 2 and 3 draw on the tunnel-diode lecture slides used in earlier years of this course (third-party material, now archived).

Figures. All figures were made for KKKC6353 (scripts in `figures-source/`). The RTD transmission and current were calculated exactly for the structure shown.